# Ziwen Yang

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## Education

**Shanghai Jiao Tong University** Sep 2025–Present

M.S. student in Logistics Engineering and Management · Shanghai, China

**Shanghai Jiao Tong University** Sep 2021–Jun 2025

B.Eng. in Industrial Engineering · Shanghai, China

Undergraduate thesis · *Yard Crane Scheduling Method in Automated Container Terminals* · Supervisor Wei Qin

**KTH Royal Institute of Technology** Jan–Jun 2024

Exchange studies in Industrial Engineering · Stockholm, Sweden

## Research Interests

Operations research for electrified transportation and energy systems, with an emphasis on vehicle-to-grid scheduling and pricing. Methods include mathematical programming, stochastic simulation optimization, discrete-choice modeling, and reinforcement learning.

## Peer-Reviewed Publication

Yang, J., Ma, Y., **Yang, Z.**, Zhou, L., and Geng, N. (2026). Dynamic Charging and Discharging Scheduling of Electric Vehicles Based on Deep Reinforcement Learning Algorithm. *Industrial Engineering Journal*, 29(5), 48–58. DOI: 10.3969/j.issn.1007-7375.250130. (In Chinese.)

## Manuscript Under Review

**Yang, Z.**, Zhou, L., and Geng, N. Choice-based stochastic pricing for vehicle-to-grid-enabled electric vehicle charging stations under spatial competition and frequency regulation commitments. Under review at *Transportation Research Part E*.

## Research Experience

**Vehicle-to-Grid Scheduling for Electric Bus Fleets** Ongoing

*Shanghai Jiao Tong University*

- Formulated a model that jointly schedules bus duties and bidirectional charging while accounting for service reliability and battery degradation.
- Implemented a branch-and-price prototype in Python with COPT and built benchmark instances from public timetables covering 5, 20, and 50 routes in Shanghai's Songjiang district.

**Day-Ahead Pricing and Grid Services at V2G Charging Stations** Oct 2025–Apr 2026

*Shanghai Jiao Tong University · First-author study*

- Used nested logit to model drivers' charging, discharging, and station choices. Embedded the model in a Monte Carlo framework that jointly optimizes tariffs, discharge compensation, power allocation, and regulation commitments.
- Developed a surrogate-assisted search and validated the selected policies with high-precision simulation. The surrogate reduced runtime relative to direct Monte Carlo evaluation in numerical experiments.

## Research Experience Continued

**Real-Time EV Charging and Discharging Dispatch** Mar–Jul 2025

*Shanghai Jiao Tong University · Collaboration with State Grid Shanghai*

- Developed an event-triggered dispatch policy using proximal policy optimization (PPO) to respond to stochastic vehicle arrivals and departures at a commercial charging station.
- Evaluated the policy against benchmark algorithms across station scales, user mixes, and tariffs. The published study found higher aggregator revenue in the tested scenarios.

**Yard-Crane Scheduling at an Automated Container Terminal** Jan–Jun 2025

*Shanghai Jiao Tong University · Industry collaboration*

- Led the yard-crane scheduling subproject. Modeled dynamically arriving tasks as a Markov decision process and implemented a pointer-network policy trained with PPO.
- Integrated the scheduler into an enterprise digital twin and evaluated operating time and crane idleness against historical schedules.

## Industry Experience

**Algorithm Development Intern** Jun–Sep 2026

*International e-commerce supply chain and logistics*

- Formulated a mixed-integer linear program for line-haul vehicle scheduling with multi-stop pickup and delivery, slot capacity, service deadlines, and inventory costs.
- Used OR-Tools and SCIP to solve rolling-horizon subproblems and built visual diagnostics to trace dispatch decisions and tune model parameters.

## Selected Honors and Awards

**Grand Prize** Oct 2023

Tsinghua IE Liangjian National Industrial Engineering Application Case Competition

Team project on post-disaster facility location and resource allocation using two-stage optimization, Benders decomposition, and simulation.

**First Prize** Oct 2023

China College Students Industrial Engineering and Lean Management Innovation Competition

Team project on automated guided vehicle path planning at a large container terminal.

**University Honors**

Undergraduate Outstanding Scholarship, Changjiang Siyuan Science and Technology Innovation Scholarship, and Outstanding Communist Youth League Member.

## Skills and Languages

**Optimization** · COPT, OR-Tools, SCIP

**Programming and research tools** · Python, PyTorch, NumPy, pandas, C++, MATLAB, Git, LaTeX

**Languages** · Chinese (native), English